

Swapnil Pant

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EDUCATION

Master of Science in Financial Technology & Analytics – Concentration: *Financial Data Science* Sep 2024 – May 2026
Stevens Institute of Technology, School of Business, Hoboken, NJ GPA: 3.97/ 4.00

Relevant Courses: Applied Statistics with App in Finance, Time Series Analysis, Risk Management

Bachelor of Technology in Computer Science & Engineering Oct 2020 – May 2024
Jaypee University of Engineering & Technology, India

EXPERIENCE

Quantitative Analyst Jul 2025 – Aug 2025
Titans of Wall Street Academy Hoboken, NJ

- Built and tested performance persistence algorithms across hedge funds using the Lipper Hedge Fund Database, revealing a near-linear decline in "performance chasing" effectiveness from 1990 to the present.
- Conducted hedge fund performance analysis using Python by calculating Sharpe Ratio, Calmar Ratio, and Annualized Volatility, while examining the impact of survivorship bias on return distributions.
- Designed and backtested a 200-day moving average strategy on the S&P 500 (1991–2025), achieving a Sharpe ratio of 1.3 and confirming normally distributed returns through visualization and statistical testing.
- Evaluated multiple moving average configurations (50d, 100d, 200d, 300d) and selected the most effective based on historical performance and drawdown characteristics.

Graduate Assistant May 2025 – May 2026
School of Business, Stevens Institute of Technology Hoboken, NJ

- Maintained student/alumni database (200+ records) and semester program data using Salesforce CRM and Excel.
- Assisted with data preparation and quality checks for assessments and performance tracking in finance and analytics courses.

React Developer Intern Jun 2023 – Aug 2023
Gonzo Digital India

- Built and deployed three data-driven web applications in React.js, collaborating across design and product teams to increase user engagement by 20%.
- Executed over 200 performance tests and resolved 50+ bugs, enhancing system reliability by 30%.

ACADEMIC PROJECTS

RegimeSense | R Oct 2025 – Dec 2025

- Developed a market regime detection framework using K-means clustering and Hidden Markov Models (HMM) to identify structural shifts in financial markets.
- Applied findings to a dynamic multi-asset allocation strategy, improving downside risk versus SPY buy-and-hold benchmarks.
- Designed a 3-state portfolio allocation (Calm/Neutral/Turbulent) using HMM regime signals, reducing drawdowns during crisis periods relative to SPY buy-and-hold.

Predicting Stock-Level Risk Premiums | Python Oct 2024 – Dec 2024

- Built and evaluated predictive models (Linear Regression, XGBoost, LightGBM, Neural Networks) on a 4M-observation panel dataset spanning 32,793 firms (1957–2021) to forecast stock-level risk premiums.
- Engineered and selected features from 94 firm-specific and 8 macroeconomic predictors, addressing multicollinearity and missingness through imputation and correlation-based filtering.
- Applied SHAP values and correlation analysis to identify key return drivers, with neural networks achieving the lowest validation RMSE among tested models.
- Constructed a top 100 portfolio from model predictions, closely tracking the -3.42% market benchmark with reduced volatility (4.75%), demonstrating that improved predictive accuracy did not translate to outperformance consistent with market efficiency.

SKILLS

Programming: Python (pandas, NumPy, scikit-learn), SQL, R (tidyverse), C++, Power BI

Data & Tools: Excel (PivotTables, VLOOKUP, macros), PowerPoint, Salesforce CRM, Bloomberg Terminal, Git/GitHub

Modeling & Methods: Regression, Classification, Tree-based models (XGBoost), PCA, Hidden Markov Models (HMM)

Analytics & Finance: Risk modeling, Time-series analysis, Anomaly detection, Forecasting, Data Visualization

Certifications: Machine Learning in Finance (*Stevens*), Bloomberg Market Concepts (*BMC*)

HONOURS & ACHIEVEMENTS

- Outstanding Leadership & Service Award, MS in Financial Analytics
- Beta Gamma Sigma (BGS) - Top 20% of graduate program